Table 4: Overview of Notations

f, f'	Neural networks
$f^{(l)}, f'^{(l')}$	Layer l/l' of neural networks f, f'
L, L'	Total number of layers in f, f'
D, D'	Number of neurons of a layer
N	Number of inputs
C	Number of classes in a classification task
m	Similarity measure
q	Quality function
$\boldsymbol{y}$	Vector of ground-truth labels
$\boldsymbol{X}$	$N \times p$ matrix of N inputs
$\boldsymbol{R}, \boldsymbol{R'}$	$N \times D / N \times D'$ representation matrices
$\boldsymbol{O}, \boldsymbol{O'}$	$N \times C$ output matrices
$\boldsymbol{S}, \boldsymbol{S'}$	Representational Similarity Matrices (RSMs)
$\mathcal{R}, \mathcal{O}$	Sets of representation/output matrices
$\mathcal{T}$	Group of linear transformations
$\sim_{\mathcal{T}}$	Equivalence up to transformations from $\mathcal{T}$
$\mathrm{O}(D)$	Group of orthogonal transformations
$\mathrm{GL}(D, \mathbb{R})$	Group of invertible matrices in $\mathbb{R}^{D \times D}$
$\mathcal{N}^k_{\boldsymbol{R}}(i)$	Set of k nearest neighbors of i in $\boldsymbol{R}$
T	Transpose of a matrix/vector
$\mathbf{1}_n$	Vector of n ones
$\boldsymbol{I}_n$	Identity matrix of size $n \times n$
$\boldsymbol{H}_n$	Centering matrix of size $n \times n$
$\mathbb{1}$	Indicator function

A Overview of Notations and Basic Definitions

A.1 Notations

Within the notations in this survey, we use a few conventions. Sets are usually denoted with uppercase calligraphic letters, such as $\mathcal{N}, \mathcal{O}, \mathcal{P}$. Matrices $\boldsymbol{M} \in \mathbb{R}^{n_1 \times n_2}$, $n_1, n_2 \in \mathbb{N}$ are always denoted with bold uppercase letters, whereas vectors $\boldsymbol{v} \in \mathbb{R}^n, n \in \mathbb{N}$ are denoted with bold lowercase letters. General scalar variables $a \in \mathbb{R}$ are usually denoted with regular lower-case letters, whereas specific constants, such as the number of classes C in a classification task, or the dimension of representations D, are denoted with upper-case letters. Specific lower-case variables are reserved, such as m for model similarity measures, or f for layer functions of neural networks. All of these fixed variables are given in Table 4, all other variables are excluded there.

A.2 Norms and Inner Products for Matrices

Here, we briefly describe the Frobenius and nuclear norm that are used in some representational similarity measures.

Frobenius Norm. On the vector space of all matrices in $\mathbb{R}^{n \times d}$, $n, d \in \mathbb{N}$, the *Frobenius inner product* is defined as

$$\langle \boldsymbol{A}, \boldsymbol{B} \rangle_F = \sum_{i=1}^n \sum_{j=1}^d \boldsymbol{A}_{i,j} \boldsymbol{B}_{i,j} = \mathrm{tr}(A^\mathsf{T} B). \tag{A.1}$$

This inner product induces the *Frobenius norm*, which for $A \in \mathbb{R}^{n \times d}$ is defined as

$$\|\boldsymbol{A}\|_F = \sqrt{\langle \boldsymbol{A}, \boldsymbol{A} \rangle_F} = \sqrt{\sum_{i=1}^n \sum_{j=1}^d |\boldsymbol{A}_{i,j}|^2} = \sqrt{\mathrm{tr}(A^\mathsf{T} A)} = \sqrt{\sum_{i=1}^{\min(n,d)} \sigma_i^2}, \tag{A.2}$$

with σ_i denoting the i-th singular value of A. The Frobenius norm is invariant to orthogonal transformations.

Nuclear Norm. Similar to the Frobenius norm, one can define the nuclear norm in terms of the singular values of a matrix

$$\|\boldsymbol{A}\|_* = \sum_{i=1}^{\min(n,d)} \sigma_i. \tag{A.3}$$

A.3 Similarity Functions for RSMs

When analyzing representational similarity, instance-wise similarity functions $s : \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}$, $n \in \mathbb{N}$ are often needed, in particular for RSM-based measures (here $n = D$). As noted in Section 3.3, they further strongly impact

Table 5: Overview of Instance-wise similarity functions

	Induced Invariances						
Function	PT	OT	IS	ILT	TR	AT	Metric
Euclidean distance	✓	✓	✗	✗	✓	✗	✓
Cosine similarity	✓	✓	✓	✗	✗	✗	✗
Linear kernel	✓	✓	✗	✗	✗	✗	✗
RBF kernel	✓	✓	✗	✗	✓	✗	✗
Pearson correlation	✓	✓	✓	✗	✓	✗	✗

which groups of transformations these measures are invariant to. In the following, we provide a brief overview of common similarity functions, where we always assume two input vectors $v, v' \in \mathbb{R}^n$ to be given. We also provide an overview of the invariances that they induce on RSM-based measures in Table 5.

- **Euclidean Distance.** This well-known distance function is defined as
$$\|v - v'\|_2 = \sqrt{\sum_{i=1}^n (v_i - v_i')^2} \tag{A.4}$$
This function satisfies the properties of a distance metric.
- **Cosine Similarity.** The cosine similarity between two vectors is defined as
$$\text{cos-sim}(\boldsymbol{v}, \boldsymbol{v}') = \frac{\boldsymbol{v}^\top \boldsymbol{v}'}{\|\boldsymbol{v}\|_2 \|\boldsymbol{v}'\|_2}. \tag{A.5}$$
It is bounded in the interval $[-1, 1]$, with $\text{cos-sim}(\boldsymbol{v}, \boldsymbol{v}') = 1$ indicating that both vectors point in the exact same direction, and $\text{cos-sim}(\boldsymbol{v}, \boldsymbol{v}') = 0$ indicating orthogonality.
- **Linear Kernel.** This kernel is defined as
$$K(\boldsymbol{v}, \boldsymbol{v}') = \boldsymbol{v}^\top \boldsymbol{v}'. \tag{A.6}$$
When $\boldsymbol{v}, \boldsymbol{v}$ have unit norm, this measure is equivalent to cosine similarity. $K(\boldsymbol{v}, \boldsymbol{v}') = 0$ indicates that vectors are orthogonal to each other. The linear kernel is not bounded.
- **Radial Basis Function Kernel.** The radial basis function (RBF) kernel is defined as
$$K_\sigma(\boldsymbol{v}, \boldsymbol{v}') = \exp\Big(-\frac{\|v - v'\|_2^2}{2\sigma^2}\Big), \tag{A.7}$$
where $\sigma \in \mathbb{R}$ is a free parameter. With a range of $[0, 1]$, $K_\sigma(\boldsymbol{v}, \boldsymbol{v}') = 1$ indicates maximum similarity, a value of zero indicates minimal similarity.
- **Pearson Correlation.** Letting $\bar{\boldsymbol{v}} = \frac{1}{n}\sum_{i=1}^n \boldsymbol{v}_i$ denote the average value of the vector $\boldsymbol{v}$, the Pearson correlation coefficient is defined as
$$r(\boldsymbol{v}, \boldsymbol{v}') = \frac{\sum_{i=1}^n (\boldsymbol{v}_i - \bar{\boldsymbol{v}}_i)(\boldsymbol{v}_i' - \bar{\boldsymbol{v}}_i')}{\sqrt{\sum_{i=1}^n (\boldsymbol{v}_i - \bar{\boldsymbol{v}}_i)^2}\sqrt{\sum_{i=1}^n (\boldsymbol{v}_i' - \bar{\boldsymbol{v}}_i')^2}}. \tag{A.8}$$
This function is bounded in the interval $[-1, 1]$, with $r(\boldsymbol{v}, \boldsymbol{v}') = 1$ indicating perfect correlation, and $r(\boldsymbol{v}, \boldsymbol{v}') = 0$ no correlation at all. When $\boldsymbol{v}, \boldsymbol{v}'$ are mean-centered, i.e. $\bar{\boldsymbol{v}} = \bar{\boldsymbol{v}}' = 0$, this function is equivalent to cosine similarity.

A.4 Intertwiner Groups

Godfrey et al. [47] introduced the concept of *intertwiner groups*, which they applied to analyze symmetries in neural network models. Formally, given an invertible activation function $\sigma : \mathbb{R} \longrightarrow \mathbb{R}$, its corresponding intertwiner group is defined as:
$$\mathrm{G}_\sigma := \mathrm{G}_{\sigma,D} = \{\boldsymbol{A} \in \mathrm{GL}(D, \mathbb{R}) : \exists \boldsymbol{B} \in \mathrm{GL}(D, \mathbb{R}) \text{ s.t. } \sigma \circ \boldsymbol{A} = \boldsymbol{B} \circ \sigma\}, \tag{A.9}$$
where $\mathrm{GL}(D, \mathbb{R})$ denotes the general linear group of invertible matrices in $\mathbb{R}^{D \times D}$. The corresponding group of transformations of neural representations is then defined as
$$\mathcal{T}_\sigma = \{\boldsymbol{R} \mapsto \boldsymbol{R}\boldsymbol{M} : \boldsymbol{M} \in \mathrm{G}_\sigma\}. \tag{A.10}$$
We highlight the case where $\sigma = \mathrm{ReLU}$, which yields the group $\mathcal{T}_{\mathrm{ReLU}}$, because Godfrey et al. [47] detail similarity measures that are invariant to transformations from $\mathcal{T}_{\mathrm{ReLU}}$. G_{ReLU} consists of matrices of the form $\boldsymbol{P}\boldsymbol{D}$, where $\boldsymbol{P} \in \mathcal{P}$ is a permutation matrix and $\boldsymbol{D}$ is a diagonal matrix with positive elements. Thus, invariance to $\mathcal{T}_{\mathrm{ReLU}}$ implies invariance to permutations, and when assuming representations with normalized columns, one can, for instance, constrain the Orthogonal Procrustes measure to be invariant to this group [47].

B Preprocessing of Representations

Next, we discuss techniques for normalization, adjusting dimensionality, and flattening of representations.

Normalization. Some similarity measures assume that the representations are normalized. For instance, it is commonly assumed that representations are mean-centered in the columns [73, 155, 101]. Mean-centering effectively constitutes a translation of the representations, which imposes the assumption that representations are equivalent under translations. In consequence, the corresponding measures are invariant towards translations. For such reasons, normalization methods should be used with caution, as they require the compared representations to be compatible with such assumptions.

In the following, we briefly discuss some commonly used normalization methods for representations or RSMs. To keep the broader scope, we consider normalization of matrices $\boldsymbol{M} \in \mathbb{R}^{n \times d}, n, d \in \mathbb{N}$. Then, a normalization can be considered as a mapping $\psi : \mathbb{R}^{n \times d} \longrightarrow \mathbb{R}^{n \times d}$. To simplify notation, in this context we apply the *centering matrix* $\boldsymbol{H}_n$, $n \in \mathbb{N}$, which is defined as $\boldsymbol{H}_n = \boldsymbol{I}_n - \frac{1}{n}\boldsymbol{1_n}\boldsymbol{1_n}^\intercal$,

- **Rescaling of Instances to Unit Norm.** Letting $\boldsymbol{D} = \operatorname{diag}(\|\boldsymbol{M}_1\|_2, \ldots, \|\boldsymbol{M}_n\|_2)$ denote the diagonal matrix of row lengths, this rescaling can be written as a transformation
$$\boldsymbol{M} \mapsto \boldsymbol{D}^{-1}\boldsymbol{M}. \tag{B.11}$$
This transformation preserves angles but alters Euclidean distances between vectors.
- **Rescaling of Columns to Unit Norm.** Letting $\boldsymbol{D} = \operatorname{diag}(\|\boldsymbol{M}_{-,1}\|_2, \ldots, \|\boldsymbol{M}_{-,d}\|_2)$ denote the diagonal matrix of column lengths, this rescaling can be written as a transformation
$$\boldsymbol{M} \mapsto \boldsymbol{M}\boldsymbol{D}^{-1}. \tag{B.12}$$
This transformation preserves neither angles nor distances.
- **Rescaling of Matrix to Unit Norm.** This preprocessing rescales the whole matrix to unit norm:
$$\boldsymbol{M} \mapsto \frac{\boldsymbol{M}}{\|\boldsymbol{M}\|_F}. \tag{B.13}$$
Like the previous rescaling, angles are preserved, but Euclidean distances are not.
- **Mean-Centering of Columns**. This normalization sets the column means to zero, while preserving their variance. It can be written as a transformation
$$\boldsymbol{M} \mapsto \boldsymbol{H}_n\boldsymbol{M}, \tag{B.14}$$
which effectively constitutes a translation of the representations. Thus, it alters angles but preserves Euclidean distance between representations.
- **Double Mean-Centering.** This approach translates both rows and columns such that both row and column means equal zero. For any matrix $\boldsymbol{M} \in \mathbb{R}^{n \times d}, n, d \in \mathbb{N}$, double mean-centering in rows and columns can be defined as a transformation
$$\boldsymbol{M} \mapsto \boldsymbol{H}_n\boldsymbol{M}\boldsymbol{H}_d \tag{B.15}$$
This normalization is typically not applied directly to representations, as it would translate individual rows differently, and alter both Euclidean distance and angles between the row vectors.

Adjusting Dimensionality. Many of the representational similarity measures presented in Section 3 implicitly assume that the representations $\boldsymbol{R}, \boldsymbol{R}'$ have the same dimensionality, i.e., $D = D'$. Thus, if $D < D'$, some preprocessing technique must be applied to match the dimensionality. Two techniques have been recommended for preprocessing: zero-padding and dimensionality reduction, such as principal component analysis (PCA) [183, 155]. When zero-padding, the dimension D of representation $\boldsymbol{R}$ is inflated by appending $D' - D$ columns of zeros to $\boldsymbol{R}$. PCA conversely reduces the dimension of the representation $\boldsymbol{R}'$ by removing the $D' - D$ lowest-information components from the representation.

Flattening. Representational similarity measures assume matrices $\boldsymbol{R} \in \mathbb{R}^{N \times D}$ as input. However, some models such as convolutional neural networks (CNNs) produce representations of more than two dimensions, making them incompatible with these measures. In such a case, representations have to be flattened, taking into account model-specific properties of representations. For example, representations from CNNs usually have the form $\boldsymbol{R} \in \mathbb{R}^{N \times h \times w \times c}$, where h, w denote height and width of the feature maps, and c the number of channels. Directly flattening these representations into matrices $\boldsymbol{R} \in \mathbb{R}^{N \times hwc}$ would yield a format in which permuting the features would disregard the spatial information in the original feature map, which may be undesirable. To avoid this issue, flattening CNN representations into matrices $\boldsymbol{R} \in \mathbb{R}^{Nhw \times c}$ yields representations where permutations only affect the channels [155]. However, when comparing two models f, f', their flattened representations are only compatible if the height and width of both models match or a feature map is upsampled, as the number of rows in the resulting matrices must match. Further, computational cost of a similarity measure may be affected by the new effective numbers of features and inputs in the flattened representation.